1. Introduction

For this project we investigate whether a simple trend-following rule can improve the risk-adjusted performance of a conventional balanced portfolio. Our conventional portfolio holds 60% in a global equity fund and 40% in government bonds, with the portfolio being rebalanced each month. The moving-average approach is related to the allocation model discussed by Faber [1].

We then compare this against an alternative strategy which applies a separate trend signal to the equity and bond allocations. If the price of an asset at the end of the month is above its ten-month moving average, we keep our position in the asset or re-invest the corresponding cash. If it is below the moving average, we move the relevant allocation into cash or keep it there. The signal is used at the first available market open in the following month.

We are not trying to find a guaranteed profitable investment strategy. Instead, we use a transparent and reproducible backtest to see the historical trade-off between return, volatility, drawdown risk and transaction costs.

All results were generated via python code which can be found at this github adress:

https://github.com/Louiskungfui/Investment-Portfolio-evaluation

1.1. Portfolio Returns

Let $R_{E,t}$ and $R_{B,t}$ be the monthly total returns of the equity and bond funds respectively. We can then calculate the monthly return of the static benchmark by

$$R_{P,t} = 0.6R_{E,t} + 0.4R_{B,t}.$$

If V_t is the value of the portfolio at the end of month t, we update the value using

$$V_t = V_{t-1}(1 + R_{P,t}).$$

This gives us the baseline which we use to compare the trend-following strategy against. The benchmark is rebalanced back to its target weights at the beginning of each month.

1.2. Trend-Following Rule

For each asset, let P_t be its price at the end of the month. We calculate its ten month moving average using

$$MA_t^{(10)} = \frac{1}{10}\sum_{j=0}^{9} P_{t-j}.$$

From this we can define the trading signal as

$$S_t = \begin{cases} 1, & P_t > MA_t^{(10)}, \\ 0, & P_t \leq MA_t^{(10)}. \end{cases}$$

When $S_t = 1$, we hold the relevant asset during month $t + 1$. When $S_t = 0$, we hold that allocation in cash instead. We calculate the signal using the information available at the end of month t and apply it at the first available market open in the following month.

2. Data and Methodology

2.1. Data

We use adjusted opening and closing prices for three US listed funds. Vanguard Total World Stock ETF (VT) represents global equities, iShares 7–10 Year Treasury Bond ETF (IEF) represents government bonds and SPDR Bloomberg 1–3 Month T-Bill ETF (BIL) represents cash. We download the prices from Yahoo Finance using the `yfinance` Python package. Adjusted prices are used since they account for distributions, dividends and stock splits. We also checked the fund descriptions against the information published by each provider [2, 3, 4].

We use the equity and bond funds to construct both portfolios. When the trend-following strategy moves out of either equities or bonds, we place the relevant allocation into the cash proxy. Since all three funds are USD-denominated, we give all results in US dollars. We use USD here as the data was easier to import and work with.

We download daily prices from 1 July 2008 to 30 June 2026 and convert the closing prices into end of month values. The first twelve months are kept as moving-average history, which gives us a common portfolio analysis period from 31 July 2009 to 30 June 2026. We retain the adjusted opening prices as well so that the backtest can apply a signal at the next available market open rather than assuming that the signal can be traded at the same closing price that generated it.

2.2. Monthly Returns

Let P_t be the adjusted closing price of an asset at the end of month t. We calculate the return during the month by

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

This gives us the percentage change in the value of the asset between two consecutive month-end prices. For the portfolio backtest, we use the daily adjusted opening and closing prices to separate overnight returns from intraday returns. This allows the old allocation to be held overnight and the new allocation to be applied at the following market open.

For example, if the value of an asset rises from \$100 to \$105 over a month, we get

$$R_t = \frac{105 - 100}{100} = 0.05.$$

Therefore the asset has a monthly return of 5%.

2.3. Static Portfolio

Our benchmark portfolio invests 60% in the equity fund and 40% in the bond fund. Therefore its return in month t is

$$R_{P,t} = 0.6R_{E,t} + 0.4R_{B,t},$$

where $R_{E,t}$ is the return of the equity fund and $R_{B,t}$ is the return of the bond fund.

At the beginning of each month we rebalance the portfolio back to these weights. The allocation held during the previous month is first exposed to the overnight movement into the new month. If equities have increased in value relative to bonds, some of the equity allocation is then sold and moved into bonds. This returns the portfolio to 60% equities and 40% bonds.

2.4. Trend-Following Portfolio

The trend-following portfolio uses the same 60/40 base allocation, but we adjust each part depending on its own trend signal.

If the price of the equity fund is above its ten-month moving average, we keep or re-invest the 60% equity allocation. Otherwise, we move this allocation into cash or keep it there. We then use the same method for the 40% bond allocation.

As a result, the portfolio can take one of four possible forms:

Equity signal	Bond signal	Portfolio allocation
1	1	60% equities, 40% bonds
1	0	60% equities, 40% cash
0	1	60% cash, 40% bonds
0	0	100% cash

We calculate the signals at the end of month t and use them to decide the portfolio allocation at the first available market open in month $t + 1$. This means that we do not use any information before it is available.

Let $R_{C,t}$ be the monthly return of the cash proxy. We also let $S_{E,t-1}$ and $S_{B,t-1}$ be the equity and bond signals observed at the end of the previous month. We can then calculate the return of the trend-following portfolio by

$$\begin{aligned} R_{T,t} = &\ 0.6S_{E,t-1}R_{E,t} + 0.4S_{B,t-1}R_{B,t} \\ &+ (1 - 0.6S_{E,t-1} - 0.4S_{B,t-1})\, R_{C,t}. \end{aligned}$$

The above equation combines the equity, bond and cash parts of the strategy, while also showing the one-month delay between the signal and the return it is applied to.

3. Performance Measures and Transaction Costs

We compare the static portfolio and the trend-following portfolio using several different performance measures. This is important since the portfolio with the largest final value will not always be the portfolio with the lowest risk.

3.1. Cumulative Return

Let V_t be the value of a portfolio at the end of month t. If the portfolio has a return of $R_{P,t}$ during this month, we update its value by

$$V_t = V_{t-1}(1 + R_{P,t}).$$

We start both portfolios with an initial value of \$1. This allows us to plot the growth of \$1 over the full period and compare the two strategies directly.

3.2. Compound Annual Growth Rate

The compound annual growth rate, or CAGR, is the constant annual return which would give us the same final portfolio value. We calculate this using

$$\text{CAGR} = \left(\frac{V_T}{V_0}\right)^{\frac{1}{Y}} - 1,$$

where V_0 is the initial portfolio value, V_T is the final value and Y is the number of years in our sample.

We use CAGR since it accounts for compounding. As a result, it gives us a better measure of long-run portfolio growth than simply taking the average monthly return.

3.3. Annualised Volatility

Volatility tells us how much the monthly returns of a portfolio vary over time. A portfolio with higher volatility experiences larger changes in value from one month to the next.

If σ_{monthly} is the standard deviation of the monthly returns, we calculate annualised volatility by

$$\sigma_{\text{annual}} = \sigma_{\text{monthly}}\sqrt{12}.$$

We use annualised volatility so that portfolio risk can be compared on the same yearly basis as CAGR.

3.4. Maximum Drawdown

A drawdown is the percentage fall in the value of a portfolio from its previous highest value. For each month, let

$$M_t = \max_{0 \le s \le t} V_s$$

be the highest value reached by the portfolio up to month t. We can then calculate the drawdown in month t by

$$D_t = \frac{V_t - M_t}{M_t}.$$

The maximum drawdown is the minimum, or most negative, value of D_t over the full period:

$$\text{Maximum drawdown} = \min_t D_t.$$

For example, if a portfolio reaches \$100 before falling to \$80, its drawdown is

$$\frac{80 - 100}{100} = -0.20.$$

Therefore the portfolio has a drawdown of -20%, or a drawdown magnitude of 20%. We use maximum drawdown since it shows us the largest loss an investor would have experienced before the portfolio recovered.

3.5. Sharpe Ratio

We calculate the Sharpe ratio using monthly excess returns. If $R_{P,t}$ is the return of a portfolio and $R_{C,t}$ is the return of BIL during the same month, we get

$$\text{Sharpe Ratio} = \sqrt{12}\frac{\overline{R_{P,t} - R_{C,t}}}{s(R_{P,t} - R_{C,t})},$$

where the bar gives the sample mean and s gives the sample standard deviation. We multiply by $\sqrt{12}$ to annualise the monthly ratio and do not use CAGR in the numerator.

A larger Sharpe ratio means that the portfolio produced a greater excess return for each unit of volatility taken.

3.6. Transaction Costs

We report the main comparison in Table 1 before transaction costs. We then use a separate sensitivity test to estimate how costs affect both portfolios. This gives us a matched net-of-cost comparison for the static and trend-following strategies.

If $w_{i,t}^{\text{target}}$ is the target weight of asset i in month t and $w_{i,t}^{\text{pre}}$ is its weight immediately before the new target is applied, the turnover used in our code is

$$\text{Turnover}_t = \frac{1}{2}\sum_i \left|w_{i,t}^{\text{target}} - w_{i,t}^{\text{pre}}\right|.$$

We calculate $w_{i,t}^{\text{pre}}$ after the previous allocation has been exposed to the overnight return. We use the factor $\frac{1}{2}$ so that we do not count the same trade twice, first when one asset is sold and again when another asset is bought.

Our main sensitivity assumption is 10 basis points for each unit of one-way turnover. Since 10 basis points is equal to 0.1%, the cost in month t is

$$\text{Cost}_t = 0.001 \times \text{Turnover}_t.$$

We charge the cost at the market open when the new target is applied. The return after costs is therefore calculated by

$$R_{T,t}^{\text{net}} = (1 + R_{T,t})(1 - \text{Cost}_t) - 1.$$

This method includes the turnover caused by both changes in the trend signal and the weights drifting between monthly rebalances. We still use a constant cost rate rather than separately modelling bid–ask spreads, market impact or fund-specific liquidity.

4. Backtesting Procedure and Sensitivity Check

We test both portfolios using the same monthly data and the same time period. Therefore, any difference in performance comes from the portfolio rules rather than from using different data.

4.1. Timing of the Strategy

We first calculate the trend signal using the price at the end of month t. This signal is then used to decide the portfolio allocation at the first available market open in month $t + 1$.

For example, if the equity fund ends month t below its ten-month moving average, we move the equity allocation into cash at the first available market open in the following month. We do not change the return in month t, since this return has already happened before we know the signal.

This is important as it stops us from using information before it is available. If we used a signal calculated at the end of a month to trade during the same month, our results would be unrealistic.

The allocation held at the previous close is used for the overnight movement into the new month. The new target allocation is then applied at the first available market open and is used for that day's intraday return and the rest of the month. This means that the backtest does not assume that a signal based on a closing price can be traded at that same closing price. We still do not model additional execution slippage or the bid–ask spread separately.

4.2. Transaction-Cost Sensitivity

We test both portfolios using transaction costs of

$$0, \quad 10, \quad \text{and} \quad 25 \text{ basis points.}$$

By increasing the assumed trading cost, we can see whether our main conclusion changes. We also test moving averages of 8, 10 and 12 months. The final 25% of the sample is kept as an out-of-sample period, beginning on 30 April 2022, and is not used to choose the main ten-month result.

4.3. Limitations

There are several limitations to our backtest.

Historical data. Our results only show how the strategy performed over the historical period used. They do not guarantee that the strategy will perform in the same way in the future.

Choice of funds. We use broad funds to represent equities and bonds. These funds may not perfectly represent every possible equity or bond portfolio.

Currency and taxes. The funds are USD-denominated, so the results do not include sterling exchange-rate movements, taxes or the cost of currency hedging for a UK investor.

Transaction costs and liquidity. We estimate transaction costs rather than observing them directly. In reality, the cost of trading will depend on the fund being used, the size of the trade, bid–ask spreads and market conditions at the time. The fixed cost assumptions therefore do not model market impact or every possible liquidity condition.

Statistical uncertainty. We use a 12-month moving-block bootstrap with 5,000 simulations to give a confidence interval for the difference in monthly returns. This is still only an estimate and does not remove the risk that the difference between the portfolios depends on the particular historical sample being used.

Length of the data. VT has a relatively short history, meaning that our backtest does not cover a full range of earlier market conditions. The out-of-sample period is also only one quarter of the available sample.

We keep these limitations in mind when interpreting our final results.

5. Results

We now compare the performance of the static 60/40 portfolio against the trend-following portfolio. We first compare both portfolios before transaction costs and then look at how matched transaction costs affect both strategies.

5.1. Portfolio Growth and Drawdowns

Figure 1 shows the performance of both portfolios over the full sample. The top panel shows the growth of $1 and the bottom panel shows the drawdowns experienced by each portfolio.

From the figure we can see that the static 60/40 portfolio produces the larger final value. An initial investment of $1 grows to $3.931 in the static portfolio, compared with $2.605 in the trend-following portfolio. This gives a CAGR of 8.39% for the static portfolio and 5.79% for the trend-following portfolio.

However, the trend-following portfolio experiences less risk. Its annualised volatility is 6.55%, compared with 9.16% for the static portfolio. We see the largest difference in the maximum drawdown: the static portfolio has a maximum drawdown of −21.30%, while the trend-following portfolio has a maximum drawdown of −8.42%. The worst full calendar-year return is also less negative for the trend-following portfolio.

The trend-following portfolio has a Sharpe ratio of 0.699, compared with 0.788 for the static portfolio. Its annualised turnover is also higher at 206.34%, compared with 35.56% for the static portfolio, and it holds an average of 26.52% in cash. Therefore, although the trend-following rule reduces volatility and drawdown risk, it does not produce a higher return or Sharpe ratio over our sample period.

The full results are shown in Table 1.

5.2. Transaction Cost Sensitivity

Table 2 shows what happens to both portfolios under different transaction-cost assumptions. The results with no transaction costs are the same as the gross results in Table 1.

As the transaction costs increase, both final values and CAGRs fall. The trend-following final value drops from $2.605 with no costs to $2.386 when we use a cost of 25 basis points.

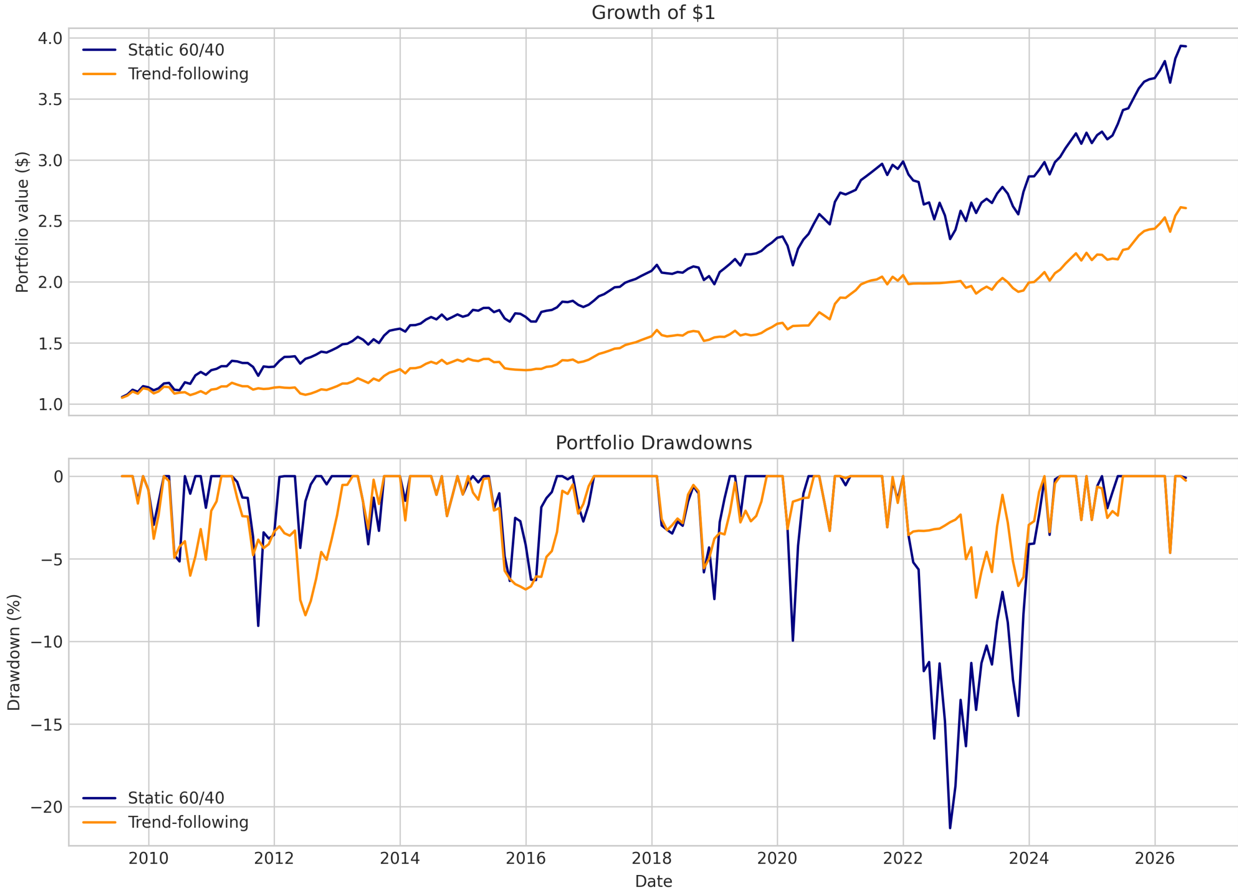

Figure 1: Growth and Drawdowns

Table 1: Full-sample performance of the two portfolios before transaction costs.

	Static 60/40 portfolio	Trend-following portfolio
Final value of $1	3.931	2.605
CAGR	8.39%	5.79%
Annualised volatility	9.16%	6.55%
Sharpe ratio	0.788	0.699
Maximum drawdown	−21.30%	−8.42%
Worst full calendar year	−16.35%	−5.23%
Annualised turnover	35.56%	206.34%
Average cash weight	0.00%	26.52%

Table 2: Matched transaction-cost comparison for the two portfolios.

Transaction costs	Portfolio	Final value of $1	CAGR	Sharpe ratio	Maximum drawdown
0 bps	Static 60/40	3.931	8.39%	0.788	−21.30%
0 bps	Trend-following	2.605	5.79%	0.699	−8.42%
10 bps	Static 60/40	3.908	8.35%	0.785	−21.33%
10 bps	Trend-following	2.515	5.57%	0.667	−8.65%
25 bps	Static 60/40	3.872	8.29%	0.779	−21.37%
25 bps	Trend-following	2.386	5.25%	0.619	−9.01%

The static portfolio also falls from \$3.931 to \$3.872. We also see the trend-following Sharpe ratio fall from 0.699 to 0.619.

The maximum drawdown becomes slightly larger for both portfolios as we increase the transaction costs. However, the trend-following maximum drawdown remains much less negative than the static 60/40 maximum drawdown. Therefore our main conclusion does not change under the transaction-cost assumptions tested. The trend-following strategy still reduces drawdown risk, but it produces less long-run growth than the static portfolio.

5.3. Out-of-Sample and Robustness Results

We keep the final 25% of the sample as an out-of-sample period. This starts on 30 April 2022 and contains 51 monthly observations. The static portfolio has a final value of \$1.395 and a CAGR of 8.14% over this period, compared with \$1.311 and 6.58% for the trend-following portfolio. The trend-following portfolio has lower volatility and a smaller maximum drawdown, but it still does not produce a higher Sharpe ratio in the holdout period.

Table 3: Performance during the final 25% out-of-sample period.

	Static 60/40 portfolio	Trend-following portfolio
Final value of \$1	1.395	1.311
CAGR	8.14%	6.58%
Annualised volatility	11.59%	7.12%
Sharpe ratio	0.391	0.369
Maximum drawdown	−16.59%	−5.58%

We also test moving averages of 8, 10 and 12 months. The results are shown below. The 10-month rule is not the best result for every measure, which reduces the concern that the conclusion is only caused by choosing the best moving-average length after looking at the full sample.

Table 4: Moving-average sensitivity for the trend-following portfolio.

Moving average	Full CAGR	Full Sharpe	Full maximum drawdown	10 bps CAGR	Out-of-sample CAGR	Out-of-sample Sharpe
8 months	6.12%	0.755	−7.61%	5.88%	6.06%	0.318
10 months	5.79%	0.699	−8.42%	5.57%	6.58%	0.369
12 months	5.78%	0.721	−7.32%	5.59%	9.01%	0.714

Finally, we use a 12-month moving-block bootstrap with 5,000 simulations. The 95% interval for the difference between the trend-following and static monthly returns is approximately $[-0.00424, 0.00004]$. Since this interval includes zero, the full-sample difference does not give strong evidence that the trend-following portfolio has a higher monthly return.

6. Conclusion

In this project we compared a static 60/40 portfolio against a trend-following portfolio using a ten-month moving average. From our results, the static portfolio produces the higher final value, CAGR and Sharpe ratio over the full sample. The same ordering remains in the final 25% out-of-sample period.

The main advantage of the trend-following portfolio is its lower downside risk. It has lower annualised volatility, a smaller worst full calendar-year loss and a considerably smaller maximum drawdown. This gives a smoother investment experience, although it comes at the cost of lower long-run growth and higher turnover.

Transaction costs reduce the performance of both portfolios further, but do not change our main result. Across the moving-average tests and the out-of-sample period, the trend-following rule continues to show lower drawdowns but not higher return or risk-adjusted performance. Overall, in this historical experiment the rule helps to limit large losses, but it does not outperform the static 60/40 portfolio in terms of either total return or risk-adjusted return.

7. Reproducibility and AI Assistance

The Python code used to download the data, construct the portfolios and create the tables and figures is available in the GitHub repository. The notebook also saves the adjusted opening and closing prices, the moving-average robustness table and the two figures used in the analysis. A requirements file and a short README explain how to run the analysis again.

Generative AI was used as a supporting tool when I ran into bugs in the Python code, when my coding knowledge was not enough to write a particular part and to make the drafting, formatting and checking process quicker. I remained responsible for checking the code, calculations, methodology and final conclusions included in the report.

References

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